17 P and Q are two different ideal gases contained in two separate identical vessels and the masses of their molecules are M_P and M_Q respectively.

The ratio of their thermodynamic temperatures, T_P and T_Q, and that of the root-mean-square speeds of their molecules, c_P and c_Q, are

$$\frac{T_P}{T_Q} = 2 \text{ and } \frac{c_P}{c_Q} = 3 .$$

What is the value of $\frac{M_P}{M_Q}$?

A $\frac{1}{6}$ **B** $\frac{2}{9}$ **C** $\frac{2}{3}$ **D** $\frac{9}{2}$